**Instructions:**

1. This question paper contains a total of **1** page (**1** side of paper).
2. Write your name, roll number, department, and section on every side of every sheet of this booklet
3. Write final answers neatly with a blue/black pen in the given boxes.
4. **Answers written outside the box will NOT be graded.**

Total **10 Marks**

**Q. 1:** Write the output of the following program in the appropriate box and answer the question. (2+2+2 = 6 Marks)

```
#include<stdio.h>
void func1(int *arr){
  for(int i=1; i<=3;i++)
    (i-1)[arr] = (i%3);
}
void func2 (int a[1] , int arr){
  a[0] = arr + 1;
}
int main(){
  int arr[4] = {0,0,0,0};
  func1(arr+1);
  for(int i=0; i<4; i++)
        printf("%d ", i[arr] );
  printf("\n");
  func2(arr, *arr);
  for(int i=0; i<4; i++)
        printf("%d ", i[arr] );
  return 0;
}
```

Output

0 1 2 0

1 1 2 0

Will the answer change if we write func2 as:

```
void func2 (int a[1] , int arr){
 int temp = arr + 1;
 a = &temp;
}
```

Explain your answer

Answer will change to
0120
0120
as a = &temp will refer to 1st location in temp array, not the value stored in the temp which was 1.

**Q. 2:** Write the output of the following sequence of instructions **long arr2[100]; int arr1[100], a1=(&arr2[50]), a2=(&arr2[10]), b1=(&arr1[50]), b2=(&arr1[10]); printf("%d %d", a1-a2 , b1-b2);** Explain your answer. (2+2 Marks)

40 40

$a_1 = 49$, $a_2 = 9$ $\therefore a_1 - a_2 = 40$

$b_1 = 49$, $b_2 = 9$ $\therefore b_1 - b_2 = 40$

& arr1[50] would refer to 50th location from beginning $\therefore$ it is 49.